



Genome structure and genetic diversity in the Ebola virus

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Abstract

Ebola is a deadly pathogen responsible for Ebola virus disease, first came to prominence in the year 1976. This rapidly evolving virus imposed a serious threat to the human population in the last few decades and also continues to be a probable threat to our race. A better understanding of the virus in terms of its genomic structure is very much needed to develop an effective antiviral therapy against this deadly pathogen. Complete knowledge of its genomic structure and variations will help us and the entire scientific community to design effective therapy in terms of either vaccine development or the development of proper antiviral medicine.

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Introduction

Viruses are not new to us and the growing hint of menaces associated with the new or re-emerging viral infections in the last few decades imposed a serious problem to humans, and therefore, we should be more cautious and scientifically informed to deal in a better way with any or every kind of viral outbreaks. Uncontrolled population growth, poor sanitation practices, and less information often play a vital role in the rapid spreading of infection during an epidemic and sometimes make the situation too worse to be controlled or contained. It has already been reported that the human population has experienced several viral epidemics or

pandemics for the last few decades, and one of them is the Ebola outbreak, which was first reported in 1976 in Central Africa and later reported in other parts of central Africa and also recently in west Africa. It is also imperative to note that the approximate spatial distribution of various pandemics caused by this particular pathogen (Ebola), reported for the last few decades confined to a geographical region stretching 10° north and south of the equator encompassing countries such as the DRC (Democratic Republic of Congo), Equatorial Guinea, Gabon, Sudan, and so on. And gradually spreading in Western African countries in recent years [1,2], there are also few reports of sporadic occurrence in other geographic locations such as the Philippines and Siena, UK and Spain [3,4].

EVD or Ebola virus disease, a type of disease characterized by viral hemorrhagic fever (VHF), is an exemplary case of zoonotic disease attributed by various anthropogenic, genetic, ecologic, socioeconomic, and climatic factors with predominant prevalence in equatorial Africa [5,6]. After transmission with the Ebola virus (EBOV), an incubation period of 2–21 days (mean 4–10) is followed by an immediate onset of Ebola hemorrhagic fever, which is characterized by fever, chills, malaise, and myalgia. The advanced stage of this particular viral infection is mainly characterized by a range of various symptoms and that include gastrointestinal (vomiting, abdominal pain, diarrhea, anorexia, nausea, etc.), respiratory (shortness of breath, cough, chest pain, nasal discharge, etc.), vascular (postural hypotension, conjunctival injection, edema, etc.), and neurological (coma, headache, confusion, etc.) conditions, which eventually results in a multiorgan failure in severe conditions. Moreover, various hemorrhagic manifestations begin to rise with the gradual progression of the disease, and in patients with fatal conditions, clinical signs are triggered near the beginning of infection and typically pass away between day 6 to day 16 with hypovolaemic shock and multiorgan failures. Interestingly all the different types of EBOV are designated as variants, as clinical manifestations of all the known types mentioned earlier do not show any kind of exclusive nature to be claimed as novel strains [1,7,8]. In this paper, we review the genome organization and genomic diversity of the known Ebola virus strains.

Ebola virus strains

It has also been known that different members of the genera Ebola virus (EBOV) belonging to the order Mononegavirales and family Filoviridae, reported causing hemorrhagic fevers with a very high mortality rate in humans with no effective treatments available for long, until very recently the first licensed vaccine for prevention of Ebola virus disease, Ervebo was developed by Public Health Agency of Canada and approved by the US FDA for use in individuals aged > 18 years [5,6,9,10]. In addition to this, it has also been found that a cocktail of three human-monoclonal antibodies (hmAbs) REGN-EB3 (INMAZEB®, Regeneron Pharmaceuticals) and another hmAb Ansuvimab (ansuvmab-zykl; EBANGA™) both of which targeted Ebola virus glycoprotein have been approved by the USFDA in 2020 as a treatment for Ebola virus infection in adults, as well as infants [11,12]. In the past outbreaks, the case fatality rates were found to be varied between 25 and 90% [13].

One of the deadliest human pathogen EBOV can be further subdivided into five species based on substantial variations in antigenicity, as well as genetic information collected, namely, Zaire Ebola virus (EBOV-Z), Sudan Ebola virus (EBOV-S), Tai Forest Ebola virus (EBOV-TF, also referred to as Cote d'Ivoire Ebola virus until 2010), Bundibugyo Ebola virus (EBOV-B), and Reston Ebola virus (EBOV-R) [13]. EBOV-Z, EBOV-S, EBOV-TF, and EBOV-B frequently cause severe hemorrhagic diseases in humans with a significantly elevated rate of case fatality. EBOV-Z and EBOV-S are the predominant strains associated with most outbreaks with greater pathogenicity than EBOV-R and EBOV-TF [14,15]. There is only one report of a non-fatal human infection caused by EBOV-TF, while EBOV-R has been reported to cause fatal infection in non-human primates [16]. Based on its high mortality rate, transmissibility, potential aerosol infectivity, and absence of effective treatment, EBOV is categorized as a BSL4 agent [17].

Almost all human cases of EVD are due to the emergence or recurrence of the EBOV-Z in regions of Gabon, the Republic of the Congo, and DRC, and of Sudan Ebola virus in Sudan and Uganda — both the species were distinct and had their first emergence in 1976 [1,18]. The preponderance of the Ebola virus in equatorial Africa is further supported by several serosurveys of selected populations from equatorial Africa over the last three to four decades that implicated that the causal agent of Ebola hemorrhagic fever is serologically cross-reactive and endemic to this geographical location [19,20]. In these endemic regions, humans, apes, and other mammalian species susceptible to this pathogenic infection are considered to be end hosts, while after

much effort naturally infected African fruit bats *Hypsignathus monstrosus*, *Epomops franqueti*, *Myonycteris torquata*, and *Rousettus aegyptiacus* were identified as reservoir species for the virus. In these reservoir species, Ebola viruses might prevail as a subclinical infection, which sometimes might be occasionally stimulated by various factors such as stress, co-infection, change, and modification in food sources, and pregnancy [1]. The disease is transmitted to humans from contact with either persistently infected non-human primates, such as the gorilla, chimpanzee, and so on, or the suspected vector host, fruit bat through episodic shedding or through a series of brief epidemics that happens to occur as the pathogen might be transmitted during the redistribution of reservoir hosts via animal migration [21]. Arbitrary human exposure to infected blood or carcasses from these hosts and adequate vulnerability of the new recipient human host are predisposing factors for the disease transmission [22]. In the disease outbreaks, most of the human infections are attributed to direct contact with infected patients or cadavers. Human-to-human transmission of EBOV is mediated through contact of infected body fluids (blood, sweat, urine, emesis, semen, breast milk, and so on.) into mucosa or skin breaks.

EBOV-Z is reported as one of the most pathogenic Ebola viruses responsible for the largest number of outbreaks with the largest number of cases. The major EBOV outbreaks with hundred or more cases that were reported to occur in Yambuku, Zaire (1976), Kikwit, Zaire (1995), the Republic of the Congo and Gabon (2001–2002), Lossi, ROC (2003), Bamoukamba, DRC (2007), and West Africa (Guinea, Sierra Leone, Liberia) in 2014–2015 have been elaborated in Table 1. It is also noteworthy to mention that a smaller outbreak also occurred in Ikanamongo, DRC in 2014 as a related event with the West African outbreak. EBOV-S was mainly identified during the major viral outbreak in Nzara Sudan in June 1976 and identified as the main causative agent for EVD in that particular region. Another noteworthy EBOV-S-induced EVD outbreak was reported in Gulu, Uganda, in 2000–2001. EBOV-B caused two outbreaks, a major one in Kabango, Uganda, in 2007 and a lesser one in Isiro, DRC, in 2012. The only known case of infection in humans by EBOV-TF was reported in course of an epizootic viral hemorrhagic fever event among western chimpanzees in the Tai National Park, Côte d'Ivoire. A comprehensive account of all identified EBOV outbreaks has been elaborated in Table 1 [1].

It is important to mention here that family Filoviridae also consist of two other genera, Marburgvirus and Cuevavirus, and it has also been reported that a high degree of similarity present in the genomic structure of

Table 1

Incidences of Ebola virus hemorrhagic fever and genome information of virus involved.

Species	Strain	Date of occurrence	Country/Location	Number of cases	Number of deaths	% Fatality	Genbank No.	Information about genome/gene	Reference
EBOV-Z	Mayinga	1976	Democratic Republic of Congo	318	280	88	NC_002549	18959bp	[34]
EBOV-Z	Eckron-76	1976	Gabon	–	–	–	U81161	GP and sGP gene (2408 bp)	[35]
EBOV-S	Boneface	1976	Sudan	284	151	53	FJ968794	18875 bp	[4]
EBOV-Z	Mayinga	1977	Democratic Republic of Congo	1	1	100	–	–	[36]
EBOV-S	Maleo	1979	Sudan	34	22	65	U23069	GP gene (2362 bp)	[37]
EBOV-R	Philippines	1989, 1992	USA	141	0	0	NC_004161	18891 bp	[38]
EBOV-R	Siena	1992	Italy	–	–	–	U23417	GP gene (2033 bp)	[37]
EBOV-R	Philippine 1992	1992	Philippines	–	–	–	U23416	GP gene (2033 bp)	[37]
EBOV-Z	Gabon 94	1994	Gabon	52	31	60	U77384	GP and sGP (GP) gene (2446 bp)	[39]
EBOV- TF	Ivory coast	1994	Ivory coast	1	0	0	U28006	GP and sGP (GP) gene (2408 bp)	[37]
EBOV-Z	Mayinga/Kikwit	1995	Democratic Republic of Congo	314	254	81	JQ352763	18944 bp	[4]
EBOV-Z	Zaire 1995	1995	Democratic Republic of Congo	–	–	–	AY354458	18961 bp	[4]
EBOV-Z	Bouee-96	1996	Gabon	91	66	73	AY058898	Zaire Ebola virus GP mRNA (2173 bp)	[40]
EBOV-R	Philippines	1996	USA	–	–	–	AB050936	18890 bp	[41]
EBOV-S	Gulu	2000	Uganda	425	224	53	AY344234	GP and sGP gene (2261 bp)	[42]
EBOV-Z	GOR Ekata	2001	Gabon	65	53	82	EU051632	GP precursor gene (2030 bp)	[43]
EBOV-Z	GOR Ekata	2001	Democratic Republic of Congo	59	44	75	EU051632	GP precursor gene (2030 bp)	[43]
EBOV-Z	GOR1 Lossi	2002	Democratic Republic of Congo	59	44	75	EU051630	GP precursor gene (2030 bp)	[43]
EBOV-Z	GOR2 Lossi	2002	Democratic Republic of Congo	59	44	75	EU051631	GP precursor gene (2030 bp)	[43]
EBOV-Z	CH Lossi	2003	Democratic Republic of Congo	178	157	88	EU051633	GP precursor gene (2030 bp)	[43]
EBOV-Z	Mandza	2003	Gabon	–	–	–	EU051635	GP precursor gene (2030 bp)	[43]
EBOV-S	Yambio	2004	.	17	7	42	EU338380	18875 bp	[4]
EBOV-Z	Etoumbi	2005	Democratic Republic of Congo	12	10	83	EU051634	GP precursor gene (2030 bp)	[43]
EBOV-B	Unknown	2007	Uganda	149	37	25	FJ217161	18940 bp	[4]
EBOV-Z	034-KS	2008	Democratic Republic of Congo	32	14	44	HQ613402	18774 bp	[4]
EBOV-R	Reston08-A	2008	Philippines	–	–	–	FJ621583	18887bp	[44]
EBOV-R	Reston08-C	2008	Philippines	–	–	–	FJ621584	18836 bp	[44]
EBOV-R	Reston08-E	2008	Philippines	–	–	–	FJ621585	18796 bp	[44]
EBOV-S	Nakissamata	2011	Uganda	1	1	100	JN638998	18875 bp	[45]
EBOV-S	EboSud-609 2012	2012	Uganda	24	17	71	KC545391	18874 bp	[46]
EBOV-S	EboSud-609 2012	2012	Uganda	7	4	57	KC545391	18874 bp	[46]
EBOV-B	EboBund-14 2012	2012	Democratic Republic of Congo	57	29	51	KC545396	18939 bp	[46]
EBOV-Z	Makona	2014–2016	Guinea	3811	2543	67	KJ660347	18959 bp	[47]
EBOV-Z	Makona	2014–2016	Liberia	10675	4809	45	KJ660347	18959 bp	[47]
EBOV-Z	Makona	2014–2016	Sierra Leone	14124	3956	28	KJ660347	18959 bp	[47]
EBOV-Z	Makona	2014	Nigeria	20	8	40	KJ660348	18959 bp	[47]
EBOV-Z	Makona	2014	Mali	8	6	75	KJ660349	18959 bp	[47]
EBOV-Z	Makona	2014	Senagal	1	0	0	KJ660350	18959 bp	[47]
EBOV-Z	Unknown	2014	USA	4	1	25	–	–	[4]
EBOV-Z	Unknown	2014	UK	1	0	0	–	–	[4]
EBOV-Z	Unknown	2014	Spain	1	0	0	–	–	[4]
EBOV-Z	Unknown	2014	Democratic Republic of Congo	66	49	74	–	99.2% similarity with JQ352763	[48]
EBOV-Z	Unknown	2015	Italy	1	0	0	–	–	[4]

(continued on next page)

Table 1. (continued)

Species	Strain	Date of occurrence	Country/Location	Number of cases	Number of deaths	% Fatality	Genbank No.	Information about genome/gene	Reference
EBOV-Z	Muyembe	2017	Democratic Republic of Congo	8	4	50	MH613311	18899 bp	[49]
EBOV-Z	Tumba-BIK	2018	Democratic Republic of Congo	54	33	61	–	variable	[49]
EBOV-Z	Ituri-MAN065	2018	Democratic Republic of Congo	142	97	68	MK007344	18938 bp	[49]

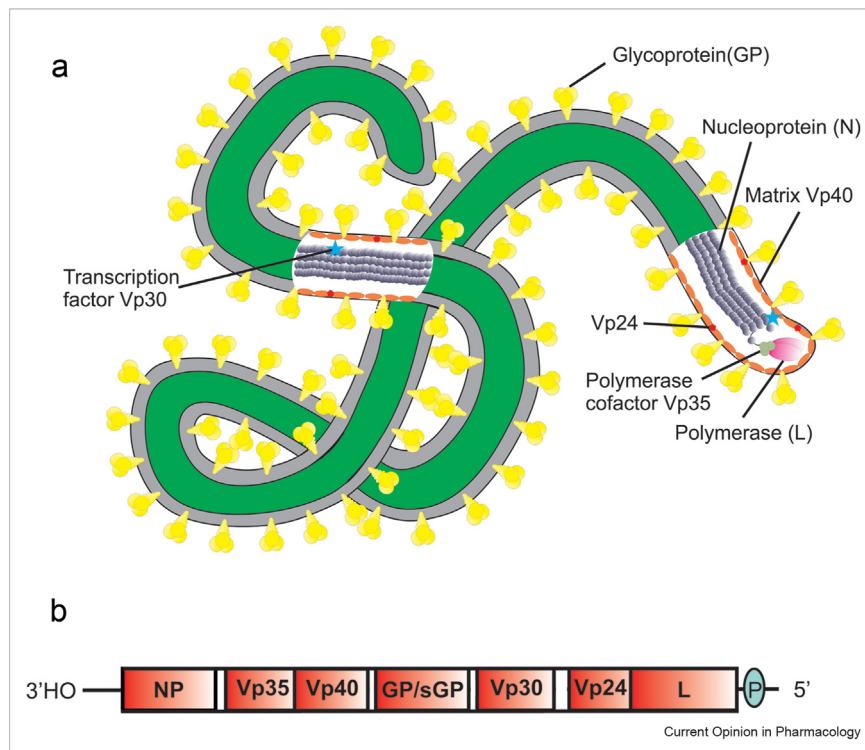
all the members belonging to family Filoviridae might indicate a possible common origin [6], and an existence of a common Marburg/Cuevavirus/Ebola-like ancestral virus might be possible between 32 and 53 million years ago by finding these endogenous viral sequences within numerous mammalian genomes [6].

Genome organization and diversity

EBOV, being a member of the Filovirus family, consists of a filamentous enveloped particle containing a non-segmented negative-strand RNA genome. The 18.9 kb EBOV genome comprises eight major subgenomic mRNAs encoding for seven structural proteins. The ribonucleoprotein complex is composed of the nucleoprotein together with viral proteins Vp24 (membrane-associated protein interfering with interferon signaling), Vp30, and Vp35 (polymerase matrix protein acting as interferon antagonist), and an RNA-dependent RNA polymerase protein (L) (Figure 1) [23,24]. The inner surface entails the matrix protein Vp40 that helps in particle formation. The surface spikes on the virion envelope are made of a trimeric transmembrane protein glycoprotein (GP) that facilitates cellular attachment and entry and is extensively glycosylated, particularly in the glycan-rich mucin-like domain [25]. The GP protein consists of GP 1 and GP 2 — two disulfide-linked furin-cleavage fragments. The Ebola virus is unique from other Mononegavirales in the production of a soluble glycoprotein (sGP) that is secreted from infected cells in large quantities [1,5,23–25].

For the identification of variant sites situated within protein-encoding regions of the Ebola virus genome, representative DNA sequences from six strains of Ebola virus retrieved from NCBI [26] were analyzed using the sequence variation analysis algorithm available within the virus pathogen database and analysis resource Ebola virus portal. Reference sequences used for the analysis were Bombali Ebola virus (NC_039345), Bundibugyo Ebola virus (NC_014373), Reston Ebola virus (NC_004161), Sudan Ebola virus (NC_006432), Tai Forest Ebola virus (NC_014372), and Zaïre Ebola virus (NC_002549). A detailed study of the whole genome data for all the six variants revealed significant alteration in nucleotide sequence encompassing all the seven genes. However, the proportion of sequence diversity was more or less the same in all the genes (Figure 2). It is noteworthy that genome sequence variation analysis from the 2014 Ebola outbreak caused by the EBOV-Z showed several nucleotide substitutions across the entire genome in a report by virus pathogen database and analysis resource [27]. Among these substitutions, a maximum number of non-synonymous alterations were observed to be concentrated in the GP gene followed by the NP gene which may attribute to the antigenic variability of the spike glycoprotein contributing to its virulence properties. The study was further elaborated

Figure 1

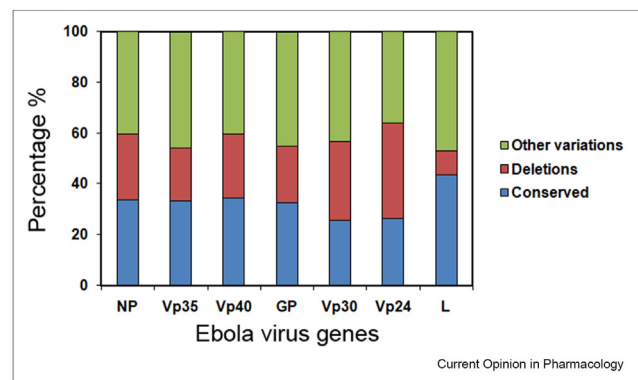


Schematic representation of structure of Ebola virus and its genomic organization. (a) The structure of Ebola virus. (b) The genome organization of Ebola virus. NP = structural nucleoprotein; Vp35 = non-structural protein; Vp40 = matrix protein; GP/sGP = envelope glycoprotein; Vp30 = non-structural protein; Vp24 = matrix protein; L = RNA polymerase.

at the levels of the seven proteins individually for all the six strains of the Ebola virus by multiple sequence alignment using algorithms of ClustalW [28]. Alignment of sequences of each of proteins from all the six strains led not only to the identification of amino acid residues that are conserved across all the strains but also the residues that undergo conservative, semi-conservative, or non-conservative replacements in different strains (Figure 3). The maximum amount of sequence conservation was noted in matrix protein Vp40 (63.7%), followed by RNA-dependent RNA polymerase protein L (61.12%). In comparison with the observation from nucleic acid sequence analysis, alignment of protein sequences showed a maximum level of sequence variation in spike glycoprotein GP (60%; of which ~36% constituted non-conservative amino acid replacements) particularly in between 300 and 500 residues of amino acid sequence. With non-conservative substitutions, biochemical features of the amino acids get changed that may be reflected in alteration of secondary/tertiary structure of the protein. Arrangement of antigenic determinants of the proteins largely depends on the conformation of the protein attributed to the domain structure and secondary/tertiary level of protein folding. So, with non-conservative replacements, the epitopes of GP proteins can be significantly altered contributing to

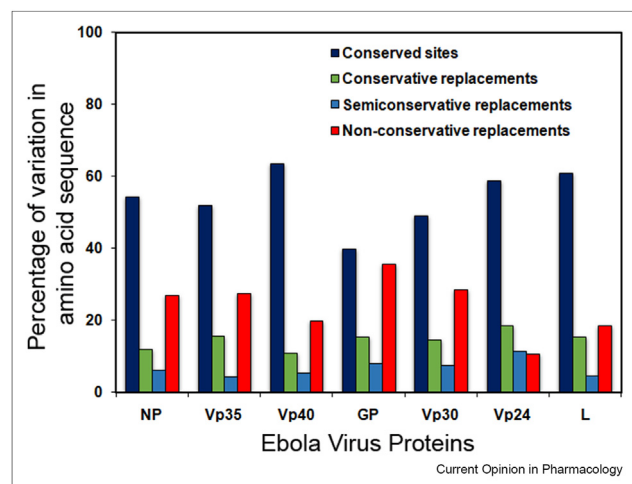
the pathogenicity of the Ebola virus [29]. A significant amount of non-conservative amino acid replacement was also noted in Vp30 (28.7%), Vp35 (27.5%), and NP (27.1%). Therefore, corroborating the nucleic acid and protein sequence analysis from all the six Ebola strains with the findings of the report using the same algorithm, we may conclude that although a large number of

Figure 2



Schematic representation of genomic variation in all strains of Ebola virus. Sequence variation analysis of five Ebola virus strains using virus pathogen database and analysis resource Ebola virus portal.

Figure 3



A graphical representation of comparative study on the distribution of amino acid sequence variations in each of the seven proteins encoded by the Ebola virus, namely, NP, Vp35, Vp40, GP, Vp30, Vp24, and L across the six strains of Ebola virus. The conserved sequence indicates the amino acids that are conserved across all the strains. Conservative replacement indicates a change of a given amino acid to a different amino acid with similar biochemical properties, semi-conservative replacement means the alteration of charge whereas non-conservative replacement indicates a change of an amino acid to another with different biochemical properties.

nucleotide diversity is distributed across the entire genome contributing to variation in amino acid sequences of the proteins, the propensity of non-synonymous changes to be sequestered in the GP locus may be responsible for the role of spike glycoprotein in pathogenicity of Ebola virus.

As mentioned earlier, Ebola is a type of RNA virus and one of the deadliest pathogens with a very higher mortality rate. As we know, that RNA viruses are highly capable of dodging various therapeutic interventions by an occurrence of a higher rate of mutation in their genome. Most of the time, it has been reported that error-prone viral RNA polymerases are evolved in such a way that they can produce a heterogeneous viral population that can easily evade different kinds of therapeutic approaches from traditional to ultra-modern [30]. To develop an effective antiviral measure and/or treatment, we need to learn the complete genomic structure of the pathogen and also various proteins secreted by the same. Generally, EBOV enters the human body either through mucous membrane or skin, and interestingly it shows a kind of wide variety of tropism to various human cells. It is imperative to mention here that cell surface lectins mainly participated in interaction with viral GPs, could be a cause of absence in the target specificity of this particular pathogen. Macrophages and dendritic cells play an important role in the early spreading of the

virus to other parts of the body because of their migratory activity and also being the first line of interaction partners of the virus [31]. This particular virus uses various methods to elude host defense mechanisms and one of them is to release different kinds of proinflammatory molecules and vasoactive modulators. It has also been reported that Vp24 and Vp35 play a crucial role in inhibiting the type I interferon responses, in particular [32]. Other viral proteins also play an important role in the pathogenesis of the virus inside the host body.

Conclusions

EVD is a deadly viral disease with a very high death rate and there was a dearth of countermeasures against the virus. However, patient survival was improved by supportive care-like rehydration with oral or intravenous fluids and symptom-specific treatments. It is also imperative to mention that a huge number of clinical trials have been conducted, and multiple therapeutics have been repurposed or designed, such as immunotherapeutics, nucleic acid-based drugs, nucleotide, and nucleoside analogs, and so on. In 2020, a co-formulated combination of three hmAbs, named REGN-EB3 (atoltivimab/maftivimab/odesivimab-ebgm, INMAZEB®) got first approval for treatment of EBOV-Z in adults and infants in the USA [11]. This was followed by the approval of an IgG1 hmAb Ansumivab (ansuvimab-zykl, EBANGA™) for the treatment of EBOV-Z infection [12]. However, the first vaccine licensed by the FDA in 2019 for preventing EVD was rVSVΔG-ZEBOV-GP Ebola vaccine (Ervebo). Rice-derived recombinant human serum albumin combined with live attenuated recombinant vesicular stomatitis virus engineered with the gene encoding the glycoprotein of EBOV-Z constitutes the Ervebo vaccine [33]. The vaccine has been reported to have an acceptable safety profile relative to the severity of Ebola virus infection and found to be effective in outbreak settings with some joint-related adverse effects like arthralgia, arthritis, etc. Also, it is mention-worthy that individuals with a history of a severe allergic reaction to rice protein should not be given Ervebo. Also because the vaccine is directed only against the GP of Zairevirus, if mutations accumulate in the glycoprotein altering the structure of the epitopes, then the efficiency of the vaccine may be compromised. Therefore, to develop a wholesome protective measure against this particular disease one needs to develop a full-proof vaccine, which will be capable of producing adequate and desirable host immune response during the infection against multiple viral epitopes in one go as seen in the case of inactivated whole organism vaccine. It is very important to have a good understanding of the viral epitopes and an idea of the entire genome structure and its variation would be considered as important prerequisites for the development of an effective vaccine. As mentioned earlier, there is a high degree of

similarity present among all reported Ebola virus genomes but variation exists in the intergenic regions and more specifically in the genes of certain proteins, such as the GP, L, and NP. Comparative genomic study may help us to design an effective vaccine that can target specific viral epitopes. This particular knowledge is very important to develop modern therapeutics to fight against this particular pathogenic agent.

Credit author statement

RPS: conceptualization, investigation, writing original draft, supervision; SG: writing, review, editing; AS: conceptualization, investigation, writing original draft; SS: investigation, writing, visualization, editing.

Conflict of interest statement

Nothing declared.

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